

Secure Multi-Branch Banking Network Infrastructure

Cisco Packet Tracer — Complete Step-by-Step Build Reference

1. Project Overview

Project Title

Secure Multi-Branch Banking Network Infrastructure

Objective

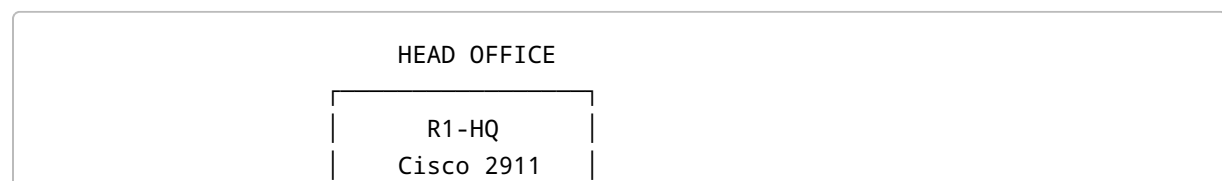
Design and implement a secure enterprise banking network connecting:

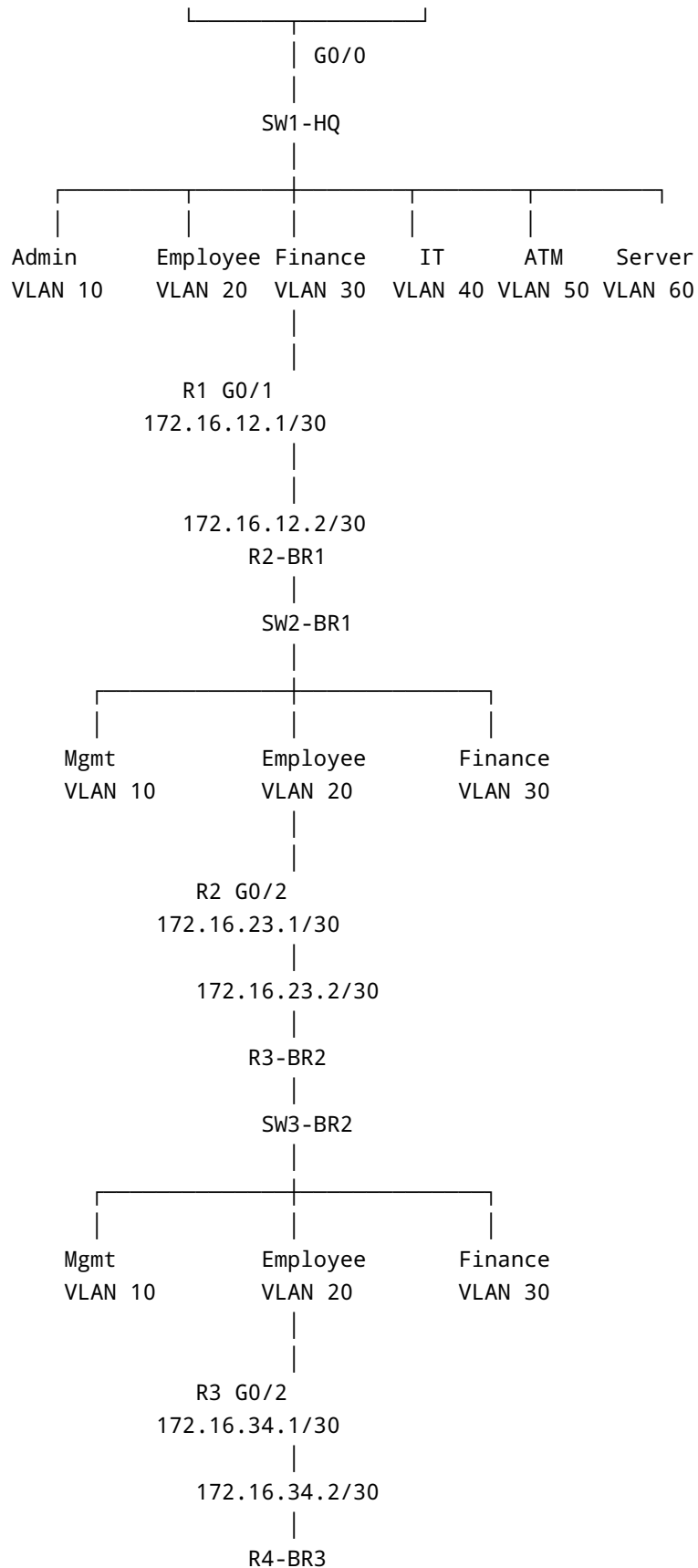
- Head Office
- Branch 1
- Branch 2
- Branch 3

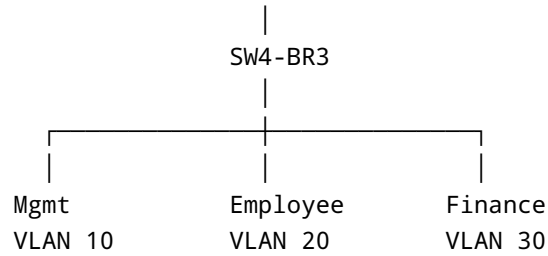
The network will demonstrate:

- VLAN
 - Inter-VLAN Routing
 - Router-on-a-Stick
 - DHCP
 - OSPF Dynamic Routing
 - WAN connectivity
 - DNS
 - HTTP Banking Server
 - ACL-based security
 - Departmental network segmentation
 - End-to-end connectivity testing
-

2. Final Network Architecture







3. Devices Required

Routers

Add:

- 4 × Cisco 2911

Rename them:

R1-HQ
R2-BRANCH1
R3-BRANCH2
R4-BRANCH3

Switches

Add:

- 4 × Cisco 2960-24TT

Rename:

SW1-HQ
SW2-BRANCH1
SW3-BRANCH2
SW4-BRANCH3

End Devices

Head Office

- PC-HQ-ADMIN

- PC-HQ-EMP
- PC-HQ-FINANCE
- PC-HQ-IT
- PC-HQ-ATM
- SERVER-BANK

Branch 1

- PC-B1-MGMT
- PC-B1-EMP
- PC-B1-FINANCE
- PC-B1-ATM

Branch 2

- PC-B2-MGMT
- PC-B2-EMP
- PC-B2-FINANCE
- PC-B2-ATM

Branch 3

- PC-B3-MGMT
- PC-B3-EMP
- PC-B3-FINANCE
- PC-B3-ATM

4. Physical Cabling

Router-to-Router

From	Port	To	Port
R1-HQ	G0/1	R2-BRANCH1	G0/1
R2-BRANCH1	G0/2	R3-BRANCH2	G0/1
R3-BRANCH2	G0/2	R4-BRANCH3	G0/1

Use Copper Cross-Over or Packet Tracer's automatic connection tool.

Router-to-Switch

Router	Port	Switch	Port
R1-HQ	G0/0	SW1-HQ	G0/1

Router	Port	Switch	Port
R2-BRANCH1	G0/0	SW2-BRANCH1	G0/1
R3-BRANCH2	G0/0	SW3-BRANCH2	G0/1
R4-BRANCH3	G0/0	SW4-BRANCH3	G0/1

Use Copper Straight-Through.

5. Head Office Switch Connections

Connect:

Device	SW1 Port	VLAN
PC-HQ-ADMIN	Fa0/1	10
PC-HQ-EMP	Fa0/2	20
PC-HQ-FINANCE	Fa0/3	30
PC-HQ-IT	Fa0/4	40
PC-HQ-ATM	Fa0/5	50
SERVER-BANK	Fa0/6	60

Router:

R1-HQ G0/0 → SW1-HQ G0/1

6. Branch 1 Switch Connections

Device	SW2 Port	VLAN
PC-B1-MGMT	Fa0/1	10
PC-B1-EMP	Fa0/2	20
PC-B1-FINANCE	Fa0/3	30
PC-B1-ATM	Fa0/4	50

Router:

R2-BRANCH1 G0/0 → SW2-BRANCH1 G0/1

7. Branch 2 Switch Connections

Device	SW3 Port	VLAN
PC-B2-MGMT	Fa0/1	10
PC-B2-EMP	Fa0/2	20
PC-B2-FINANCE	Fa0/3	30
PC-B2-ATM	Fa0/4	50

Router:

R3-BRANCH2 G0/0 → SW3-BRANCH2 G0/1

8. Branch 3 Switch Connections

Device	SW4 Port	VLAN
PC-B3-MGMT	Fa0/1	10
PC-B3-EMP	Fa0/2	20
PC-B3-FINANCE	Fa0/3	30
PC-B3-ATM	Fa0/4	50

Router:

R4-BRANCH3 G0/0 → SW4-BRANCH3 G0/1

9. IP Addressing Plan

Head Office

VLAN	Name	Network	Gateway
10	Management	10.1.10.0/24	10.1.10.1
20	Employees	10.1.20.0/24	10.1.20.1
30	Finance	10.1.30.0/24	10.1.30.1
40	IT	10.1.40.0/24	10.1.40.1
50	ATM	10.1.50.0/24	10.1.50.1
60	Servers	10.1.60.0/24	10.1.60.1

Branch 1

VLAN	Name	Network	Gateway
10	Management	10.2.10.0/24	10.2.10.1
20	Employees	10.2.20.0/24	10.2.20.1
30	Finance	10.2.30.0/24	10.2.30.1
50	ATM	10.2.50.0/24	10.2.50.1

Branch 2

VLAN	Name	Network	Gateway
10	Management	10.3.10.0/24	10.3.10.1
20	Employees	10.3.20.0/24	10.3.20.1
30	Finance	10.3.30.0/24	10.3.30.1
50	ATM	10.3.50.0/24	10.3.50.1

Branch 3

VLAN	Name	Network	Gateway
10	Management	10.4.10.0/24	10.4.10.1
20	Employees	10.4.20.0/24	10.4.20.1

VLAN	Name	Network	Gateway
30	Finance	10.4.30.0/24	10.4.30.1
50	ATM	10.4.50.0/24	10.4.50.1

10. WAN Addressing

Link	Device	Interface	IP
R1-R2	R1	G0/1	172.16.12.1/30
R1-R2	R2	G0/1	172.16.12.2/30
R2-R3	R2	G0/2	172.16.23.1/30
R2-R3	R3	G0/1	172.16.23.2/30
R3-R4	R3	G0/2	172.16.34.1/30
R3-R4	R4	G0/1	172.16.34.2/30

11. Build Order

Follow this exact order:

1. Place devices
2. Rename devices
3. Wire all devices
4. Configure switches
5. Configure router LAN interfaces
6. Configure router WAN interfaces
7. Configure OSPF
8. Configure DHCP
9. Configure server
10. Configure ACL
11. Configure PCs for DHCP
12. Test VLANs
13. Test DHCP
14. Test OSPF
15. Test inter-branch communication
16. Test server

- 17. Test ACL
- 18. Save the Packet Tracer file

Do not configure everything randomly.

12. Configure SW1-HQ

Open:

SW1-HQ → CLI

Enter:

```
enable
configure terminal

hostname SW1-HQ

vlan 10
 name MANAGEMENT
exit

vlan 20
 name EMPLOYEES
exit

vlan 30
 name FINANCE
exit

vlan 40
 name IT
exit

vlan 50
 name ATM
exit

vlan 60
 name SERVERS
exit

interface gigabitEthernet 0/1
```

```
    switchport mode trunk
exit

interface fastEthernet 0/1
    switchport mode access
    switchport access vlan 10
    spanning-tree portfast
exit

interface fastEthernet 0/2
    switchport mode access
    switchport access vlan 20
    spanning-tree portfast
exit

interface fastEthernet 0/3
    switchport mode access
    switchport access vlan 30
    spanning-tree portfast
exit

interface fastEthernet 0/4
    switchport mode access
    switchport access vlan 40
    spanning-tree portfast
exit

interface fastEthernet 0/5
    switchport mode access
    switchport access vlan 50
    spanning-tree portfast
exit

interface fastEthernet 0/6
    switchport mode access
    switchport access vlan 60
    spanning-tree portfast
exit

end
write memory
```

Check:

```
show vlan brief
show interfaces trunk
```

13. Configure SW2-BRANCH1

```
enable
configure terminal

hostname SW2-BRANCH1

vlan 10
  name MANAGEMENT
exit

vlan 20
  name EMPLOYEES
exit

vlan 30
  name FINANCE
exit

vlan 50
  name ATM
exit

interface gigabitEthernet 0/1
  switchport mode trunk
exit

interface fastEthernet 0/1
  switchport mode access
  switchport access vlan 10
  spanning-tree portfast
exit

interface fastEthernet 0/2
  switchport mode access
  switchport access vlan 20
  spanning-tree portfast
exit

interface fastEthernet 0/3
```

```
switchport mode access
switchport access vlan 30
spanning-tree portfast
exit

interface fastEthernet 0/4
switchport mode access
switchport access vlan 50
spanning-tree portfast
exit

end
write memory
```

14. Configure SW3-BRANCH2

Use the same configuration as SW2, changing:

```
hostname SW3-BRANCH2
```

Then:

```
write memory
```

15. Configure SW4-BRANCH3

Use the same configuration as SW2, changing:

```
hostname SW4-BRANCH3
```

Then:

```
write memory
```

16. Configure R1-HQ

Open:

R1-HQ → CLI

Enter:

```
enable
configure terminal

hostname R1-HQ
no ip domain-lookup

interface gigabitEthernet 0/0
  no shutdown
exit

interface gigabitEthernet 0/0.10
  encapsulation dot1Q 10
  ip address 10.1.10.1 255.255.255.0
exit

interface gigabitEthernet 0/0.20
  encapsulation dot1Q 20
  ip address 10.1.20.1 255.255.255.0
exit

interface gigabitEthernet 0/0.30
  encapsulation dot1Q 30
  ip address 10.1.30.1 255.255.255.0
exit

interface gigabitEthernet 0/0.40
  encapsulation dot1Q 40
  ip address 10.1.40.1 255.255.255.0
exit

interface gigabitEthernet 0/0.50
  encapsulation dot1Q 50
  ip address 10.1.50.1 255.255.255.0
exit

interface gigabitEthernet 0/0.60
  encapsulation dot1Q 60
```

```
ip address 10.1.60.1 255.255.255.0
exit

interface gigabitEthernet 0/1
ip address 172.16.12.1 255.255.255.252
no shutdown
exit
```

17. Configure OSPF on R1

```
router ospf 1
router-id 1.1.1.1

network 10.1.10.0 0.0.0.255 area 0
network 10.1.20.0 0.0.0.255 area 0
network 10.1.30.0 0.0.0.255 area 0
network 10.1.40.0 0.0.0.255 area 0
network 10.1.50.0 0.0.0.255 area 0
network 10.1.60.0 0.0.0.255 area 0
network 172.16.12.0 0.0.0.3 area 0

end
write memory
```

18. Configure R1 DHCP

```
enable
configure terminal

ip dhcp excluded-address 10.1.10.1 10.1.10.10

ip dhcp pool HQ-MANAGEMENT
network 10.1.10.0 255.255.255.0
default-router 10.1.10.1
dns-server 10.1.60.10
exit

ip dhcp excluded-address 10.1.20.1 10.1.20.10

ip dhcp pool HQ-EMPLOYEES
```

```
network 10.1.20.0 255.255.255.0
default-router 10.1.20.1
dns-server 10.1.60.10
exit

ip dhcp excluded-address 10.1.30.1 10.1.30.10

ip dhcp pool HQ-FINANCE
network 10.1.30.0 255.255.255.0
default-router 10.1.30.1
dns-server 10.1.60.10
exit

ip dhcp excluded-address 10.1.40.1 10.1.40.10

ip dhcp pool HQ-IT
network 10.1.40.0 255.255.255.0
default-router 10.1.40.1
dns-server 10.1.60.10
exit

ip dhcp excluded-address 10.1.50.1 10.1.50.10

ip dhcp pool HQ-ATM
network 10.1.50.0 255.255.255.0
default-router 10.1.50.1
dns-server 10.1.60.10
exit

end
write memory
```

19. Configure R2-BRANCH1

```
enable
configure terminal

hostname R2-BRANCH1
no ip domain-lookup

interface gigabitEthernet 0/0
no shutdown
exit
```

```
interface gigabitEthernet 0/0.10
  encapsulation dot1Q 10
  ip address 10.2.10.1 255.255.255.0
exit

interface gigabitEthernet 0/0.20
  encapsulation dot1Q 20
  ip address 10.2.20.1 255.255.255.0
exit

interface gigabitEthernet 0/0.30
  encapsulation dot1Q 30
  ip address 10.2.30.1 255.255.255.0
exit

interface gigabitEthernet 0/0.50
  encapsulation dot1Q 50
  ip address 10.2.50.1 255.255.255.0
exit

interface gigabitEthernet 0/1
  ip address 172.16.12.2 255.255.255.252
  no shutdown
exit

interface gigabitEthernet 0/2
  ip address 172.16.23.1 255.255.255.252
  no shutdown
exit

router ospf 1
  router-id 2.2.2.2

  network 10.2.10.0 0.0.0.255 area 0
  network 10.2.20.0 0.0.0.255 area 0
  network 10.2.30.0 0.0.0.255 area 0
  network 10.2.50.0 0.0.0.255 area 0
  network 172.16.12.0 0.0.0.3 area 0
  network 172.16.23.0 0.0.0.3 area 0

end
write memory
```


20. Configure R2 DHCP

```
enable
configure terminal

ip dhcp excluded-address 10.2.10.1 10.2.10.10

ip dhcp pool B1-MANAGEMENT
 network 10.2.10.0 255.255.255.0
 default-router 10.2.10.1
 dns-server 10.1.60.10
exit

ip dhcp excluded-address 10.2.20.1 10.2.20.10

ip dhcp pool B1-EMPLOYEES
 network 10.2.20.0 255.255.255.0
 default-router 10.2.20.1
 dns-server 10.1.60.10
exit

ip dhcp excluded-address 10.2.30.1 10.2.30.10

ip dhcp pool B1-FINANCE
 network 10.2.30.0 255.255.255.0
 default-router 10.2.30.1
 dns-server 10.1.60.10
exit

ip dhcp excluded-address 10.2.50.1 10.2.50.10

ip dhcp pool B1-ATM
 network 10.2.50.0 255.255.255.0
 default-router 10.2.50.1
 dns-server 10.1.60.10
exit

end
write memory
```

21. Configure R3-BRANCH2

```
enable
configure terminal

hostname R3-BRANCH2
no ip domain-lookup

interface gigabitEthernet 0/0
  no shutdown
exit

interface gigabitEthernet 0/0.10
  encapsulation dot1Q 10
  ip address 10.3.10.1 255.255.255.0
exit

interface gigabitEthernet 0/0.20
  encapsulation dot1Q 20
  ip address 10.3.20.1 255.255.255.0
exit

interface gigabitEthernet 0/0.30
  encapsulation dot1Q 30
  ip address 10.3.30.1 255.255.255.0
exit

interface gigabitEthernet 0/0.50
  encapsulation dot1Q 50
  ip address 10.3.50.1 255.255.255.0
exit

interface gigabitEthernet 0/1
  ip address 172.16.23.2 255.255.255.252
  no shutdown
exit

interface gigabitEthernet 0/2
  ip address 172.16.34.1 255.255.255.252
  no shutdown
exit

router ospf 1
  router-id 3.3.3.3

  network 10.3.10.0 0.0.0.255 area 0
```

```
network 10.3.20.0 0.0.0.255 area 0
network 10.3.30.0 0.0.0.255 area 0
network 10.3.50.0 0.0.0.255 area 0
network 172.16.23.0 0.0.0.3 area 0
network 172.16.34.0 0.0.0.3 area 0

end
write memory
```

22. Configure R3 DHCP

```
enable
configure terminal

ip dhcp excluded-address 10.3.10.1 10.3.10.10

ip dhcp pool B2-MANAGEMENT
network 10.3.10.0 255.255.255.0
default-router 10.3.10.1
dns-server 10.1.60.10
exit

ip dhcp excluded-address 10.3.20.1 10.3.20.10

ip dhcp pool B2-EMPLOYEES
network 10.3.20.0 255.255.255.0
default-router 10.3.20.1
dns-server 10.1.60.10
exit

ip dhcp excluded-address 10.3.30.1 10.3.30.10

ip dhcp pool B2-FINANCE
network 10.3.30.0 255.255.255.0
default-router 10.3.30.1
dns-server 10.1.60.10
exit

ip dhcp excluded-address 10.3.50.1 10.3.50.10

ip dhcp pool B2-ATM
network 10.3.50.0 255.255.255.0
default-router 10.3.50.1
dns-server 10.1.60.10
```

```
exit

end
write memory
```

23. Configure R4-BRANCH3

```
enable
configure terminal

hostname R4-BRANCH3
no ip domain-lookup

interface gigabitEthernet 0/0
  no shutdown
exit

interface gigabitEthernet 0/0.10
  encapsulation dot1Q 10
  ip address 10.4.10.1 255.255.255.0
exit

interface gigabitEthernet 0/0.20
  encapsulation dot1Q 20
  ip address 10.4.20.1 255.255.255.0
exit

interface gigabitEthernet 0/0.30
  encapsulation dot1Q 30
  ip address 10.4.30.1 255.255.255.0
exit

interface gigabitEthernet 0/0.50
  encapsulation dot1Q 50
  ip address 10.4.50.1 255.255.255.0
exit

interface gigabitEthernet 0/1
  ip address 172.16.34.2 255.255.255.252
  no shutdown
exit

router ospf 1
  router-id 4.4.4.4
```

```
network 10.4.10.0 0.0.0.255 area 0
network 10.4.20.0 0.0.0.255 area 0
network 10.4.30.0 0.0.0.255 area 0
network 10.4.50.0 0.0.0.255 area 0
network 172.16.34.0 0.0.0.3 area 0
```

```
end
write memory
```

24. Configure R4 DHCP

```
enable
configure terminal

ip dhcp excluded-address 10.4.10.1 10.4.10.10

ip dhcp pool B3-MANAGEMENT
network 10.4.10.0 255.255.255.0
default-router 10.4.10.1
dns-server 10.1.60.10
exit

ip dhcp excluded-address 10.4.20.1 10.4.20.10

ip dhcp pool B3-EMPLOYEES
network 10.4.20.0 255.255.255.0
default-router 10.4.20.1
dns-server 10.1.60.10
exit

ip dhcp excluded-address 10.4.30.1 10.4.30.10

ip dhcp pool B3-FINANCE
network 10.4.30.0 255.255.255.0
default-router 10.4.30.1
dns-server 10.1.60.10
exit

ip dhcp excluded-address 10.4.50.1 10.4.50.10

ip dhcp pool B3-ATM
network 10.4.50.0 255.255.255.0
default-router 10.4.50.1
```

```
dns-server 10.1.60.10
exit

end
write memory
```

25. Configure the Banking Server

Select:

```
SERVER-BANK
→ Desktop
→ IP Configuration
```

Set:

```
IP Address:      10.1.60.10
Subnet Mask:     255.255.255.0
Default Gateway: 10.1.60.1
DNS Server:      10.1.60.10
```

Do NOT select DHCP.

26. Enable DNS

Go:

```
SERVER-BANK
→ Services
→ DNS
```

Set:

```
DNS: ON
```

Add:

Name: bank.local
Address: 10.1.60.10

Add another record:

Name: www.bank.local
Address: 10.1.60.10

27. Enable Web Server

Go:

SERVER-BANK
→ Services
→ HTTP

Set:

HTTP: ON
HTTPS: ON

Edit the index page:

SECURE BANKING NETWORK

Welcome to National Banking Corporation

Head Office
Branch 1
Branch 2
Branch 3

Services:

- Online Banking
- ATM Services
- Account Management
- Customer Support

28. Configure All PCs for DHCP

For every PC:

```
PC
→ Desktop
→ IP Configuration
→ DHCP
```

Wait a few seconds.

The PC should receive an address.

Example:

```
IP Address:      10.1.20.11
Subnet Mask:     255.255.255.0
Default Gateway: 10.1.20.1
DNS:             10.1.60.10
```

The exact host number may differ.

29. Verify DHCP

On R1:

```
show ip dhcp binding
```

On R2:

```
show ip dhcp binding
```

On R3:

```
show ip dhcp binding
```

On R4:


```
show ip dhcp binding
```

If bindings appear, DHCP is working.

30. Configure Banking Security ACL

HQ

Employees must not access Finance.

On R1:

```
enable
configure terminal

access-list 110 deny ip 10.1.20.0 0.0.0.255 10.1.30.0 0.0.0.255
access-list 110 permit ip any any

interface gigabitEthernet 0/0.20
 ip access-group 110 in
exit

end
write memory
```

Branch 1

```
enable
configure terminal

access-list 120 deny ip 10.2.20.0 0.0.0.255 10.2.30.0 0.0.0.255
access-list 120 permit ip any any

interface gigabitEthernet 0/0.20
 ip access-group 120 in
exit

end
write memory
```

Branch 2

```
enable
configure terminal

access-list 130 deny ip 10.3.20.0 0.0.0.255 10.3.30.0 0.0.0.255
access-list 130 permit ip any any

interface gigabitEthernet 0/0.20
 ip access-group 130 in
exit

end
write memory
```

Branch 3

```
enable
configure terminal

access-list 140 deny ip 10.4.20.0 0.0.0.255 10.4.30.0 0.0.0.255
access-list 140 permit ip any any

interface gigabitEthernet 0/0.20
 ip access-group 140 in
exit

end
write memory
```

31. First Verification — Interfaces

Run on every router:

```
show ip interface brief
```

Everything important should show:

```
up    up
```

Check:

```
R1:
G0/0
G0/0.10
G0/0.20
G0/0.30
G0/0.40
G0/0.50
G0/0.60
G0/1
```

```
R2:
G0/0
G0/0.10
G0/0.20
G0/0.30
G0/0.50
G0/1
G0/2
```

```
R3:
G0/0
G0/0.10
G0/0.20
G0/0.30
G0/0.50
G0/1
G0/2
```

```
R4:
G0/0
G0/0.10
G0/0.20
G0/0.30
G0/0.50
G0/1
```

32. Second Verification — VLAN

On SW1:

```
show vlan brief
```

Expected:

```
10  MANAGEMENT
20  EMPLOYEES
30  FINANCE
40  IT
50  ATM
60  SERVERS
```

On SW2/SW3/SW4:

```
10  MANAGEMENT
20  EMPLOYEES
30  FINANCE
50  ATM
```

33. Third Verification — Trunk

On every switch:

```
show interfaces trunk
```

The router-facing:

```
G0/1
```

should be trunking.

34. Fourth Verification — OSPF

On R1:

```
show ip ospf neighbor
```

Expected neighbor:

R2

On R2:

```
show ip ospf neighbor
```

Expected:

R1
R3

On R3:

```
show ip ospf neighbor
```

Expected:

R2
R4

On R4:

```
show ip ospf neighbor
```

Expected:

R3

35. Fifth Verification — Routing Table

Run:

```
show ip route
```

Look for routes beginning with:

```
0
```

Example:

```
0 10.2.20.0/24  
0 10.3.20.0/24  
0 10.4.20.0/24
```

The `0` means the route was learned through OSPF.

36. Sixth Verification — Router Ping

From R1:

```
ping 172.16.12.2
```

Expected:

```
Success
```

From R2:

```
ping 172.16.23.2
```

Expected:

```
Success
```

From R3:

```
ping 172.16.34.2
```

Expected:

```
Success
```

37. Seventh Verification — Inter-Branch Ping

From an HQ PC, ping a Branch 1 PC.

Example:

```
ping 10.2.20.11
```

Then Branch 2:

```
ping 10.3.20.11
```

Then Branch 3:

```
ping 10.4.20.11
```

All should succeed.

38. Eighth Verification — Banking Server

From any PC:

```
ping 10.1.60.10
```

Expected:

```
Reply from 10.1.60.10
```

Then open:

```
Desktop  
→ Web Browser
```

Enter:

```
http://10.1.60.10
```

The banking webpage should load.

Then test:

```
http://bank.local
```

If DNS is working, this should also load the banking page.

39. Ninth Verification — ACL

From an HQ Employee PC:

```
ping 10.1.30.x
```

where `10.1.30.x` is the Finance PC.

Expected:

```
Request timed out
```

This is intentional.

Then:

```
ping 10.1.60.10
```

Expected:

```
Reply
```


Therefore:

```
Employee → Finance      BLOCKED
Employee → Bank Server   ALLOWED
```

40. ACL Verification

On R1:

```
show access-lists
```

You should see packet counters.

For example:

```
Extended IP access list 110
deny ip 10.1.20.0 ...
permit ip any any
```

After attempting the blocked ping, the deny counter should increase.

41. Complete Final Testing Checklist

Physical

- ☐ Four routers placed
- ☐ Four switches placed
- ☐ All PCs placed
- ☐ Banking server placed
- ☐ Router-to-router cables connected
- ☐ Router-to-switch cables connected
- ☐ PC-to-switch cables connected
- ☐ Links become green

Switches

- ☐ SW1 VLANs configured
- ☐ SW2 VLANs configured

- ☐ SW3 VLANs configured
- ☐ SW4 VLANs configured
- ☐ Access ports assigned correctly
- ☐ Router ports configured as trunks

Routers

- ☐ R1 LAN configured
- ☐ R2 LAN configured
- ☐ R3 LAN configured
- ☐ R4 LAN configured
- ☐ WAN IPs configured
- ☐ All interfaces

DHCP

- ☐ HQ Management receives IP
- ☐ HQ Employees receives IP
- ☐ HQ Finance receives IP
- ☐ HQ IT receives IP
- ☐ HQ ATM receives IP
- ☐ Branch 1 receives IP
- ☐ Branch 2 receives IP
- ☐ Branch 3 receives IP

Routing

- ☐ R1 sees R2
- ☐ R2 sees R1 and R3
- ☐ R3 sees R2 and R4
- ☐ R4 sees R3
- ☐ OSPF routes appear
- ☐ Branches can communicate

Server

- ☐ Static IP configured
- ☐ Gateway configured
- ☐ DNS enabled
- ☐ HTTP enabled
- ☐ bank.local configured
- ☐ Banking webpage loads

Security

- ☐ HQ Employee → Finance blocked
- ☐ Branch 1 Employee → Finance blocked

- [] Branch 2 Employee → Finance blocked
 - [] Branch 3 Employee → Finance blocked
 - [] Employee → Banking Server allowed
 - [] ACL counters increase
-

42. Important Troubleshooting Commands

If something does not work, don't immediately reconfigure everything.

Use these commands.

Router interfaces

```
show ip interface brief
```

Router configuration

```
show running-config
```

Routing

```
show ip route
```

OSPF

```
show ip ospf neighbor
```

DHCP

```
show ip dhcp binding  
show ip dhcp pool
```

Switch VLAN

```
show vlan brief
```

Switch trunk

```
show interfaces trunk
```

ACL

```
show access-lists
```

43. Common Problems

PC receives 169.254.x.x

DHCP failed.

Check:

```
PC → correct switch port  
Switch → correct VLAN  
Switch → trunk  
Router → correct subinterface  
Router → DHCP pool
```

Router interface says administratively down

Run:

```
configure terminal  
interface gigabitEthernet 0/0  
no shutdown
```

OSPF neighbor doesn't appear

Check both sides:

```
show ip interface brief
```

Then:

```
show ip ospf neighbor
```

Verify the WAN IP addresses and `/30` networks.

VLAN PC cannot reach gateway

Check:

```
show vlan brief  
show interfaces trunk
```

Then verify that the router has the corresponding subinterface:

```
G0/0.10  
G0/0.20  
G0/30  
...
```

and that the VLAN number matches:

```
encapsulation dot1Q 10
```

Branches cannot communicate

Check:

```
show ip ospf neighbor  
show ip route
```

If OSPF routes aren't present, check the OSPF `network` statements.

44. Final Configuration Save

After everything works, run on every router:

```
write memory
```

or:

```
copy running-config startup-config
```

Do the same for all switches.

Then save your Packet Tracer file:

```
File  
→ Save As
```

Suggested filename:

```
Secure_Multi_Branch_Banking_Network.pkt
```

45. What You Can Claim in the Project

Your project implements:

- ✓ Enterprise Banking Network
- ✓ Multi-Branch Architecture
- ✓ VLAN Segmentation
- ✓ Inter-VLAN Routing
- ✓ Router-on-a-Stick
- ✓ DHCP
- ✓ OSPF
- ✓ WAN Connectivity
- ✓ Central Banking Server
- ✓ DNS
- ✓ HTTP
- ✓ ACL Security
- ✓ Finance Department Protection

- ✓ ATM Network
- ✓ Network Monitoring/Verification

46. Recommended Demonstration Flow

During your lab presentation, demonstrate the project in this order:

Step 1 — Topology

Explain:

"Our system consists of a Head Office and three banking branches connected through routed WAN links."

Step 2 — VLAN

Show:

```
show vlan brief
```

Explain that departments are separated logically.

Step 3 — DHCP

Show:

```
show ip dhcp binding
```

Explain that clients automatically receive IP configuration.

Step 4 — OSPF

Show:

```
show ip ospf neighbor
```

Explain that OSPF dynamically exchanges routes between branches.

Step 5 — Routing

Show:

```
show ip route
```

Point out the routes.

Step 6 — Branch Communication

Ping:

```
HQ → Branch 1  
HQ → Branch 2  
HQ → Branch 3
```

Step 7 — Banking Server

Open:

```
http://bank.local
```

Show the banking webpage.

Step 8 — Security

Show:

```
Employee → Finance
```

and demonstrate that it is blocked.

Then:

```
Employee → Banking Server
```

and demonstrate that it is allowed.

47. Final Project Statement

Secure Multi-Branch Banking Network Infrastructure is an enterprise networking simulation designed in Cisco Packet Tracer to provide secure communication between a central banking headquarters and multiple branches. The system uses VLANs for departmental segmentation, router-on-a-stick for inter-VLAN

communication, DHCP for automated IP assignment, OSPF for dynamic routing, and ACLs to protect sensitive financial networks. A centralized banking server provides DNS and HTTP services for internal banking applications.

48. One-Page Quick Reference

PROJECT:
Secure Multi-Branch Banking Network Infrastructure

ROUTERS:
R1-HQ
R2-BRANCH1
R3-BRANCH2
R4-BRANCH3

SWITCHES:
SW1-HQ
SW2-BRANCH1
SW3-BRANCH2
SW4-BRANCH3

WAN:
R1-R2 = 172.16.12.0/30
R2-R3 = 172.16.23.0/30
R3-R4 = 172.16.34.0/30

HQ:
10.1.x.x

BRANCH 1:
10.2.x.x

BRANCH 2:
10.3.x.x

BRANCH 3:
10.4.x.x

VLAN:
10 = Management
20 = Employees
30 = Finance
40 = IT
50 = ATM
60 = Servers

ROUTING:

OSPF Area 0

DHCP:

Configured on each branch router

SERVER:

10.1.60.10

DNS:

bank.local

WEB:

http://bank.local

SECURITY:

Employees → Finance = BLOCKED

KEY COMMANDS:

show ip interface brief

show vlan brief

show interfaces trunk

show ip ospf neighbor

show ip route

show ip dhcp binding

show access-lists

Build principle: Physical topology → Switch VLANs → Router subinterfaces → WAN → OSPF → DHCP → Server → ACL → Testing. Follow that sequence and troubleshoot at the layer where the failure occurs.